

Question number	Answer	Additional guidance	Mark
3(a)(i)	A description that includes the following points: • compete with pathogen for {space / nutrients / metabolites / named nutrient} (1) • (skin flora) producing toxins (to pathogens) (1) • preventing {the growth of the pathogens / colonisation (of the skin) by pathogens} (1)	ACCEPT other {bacteria / microorganisms} for pathogens ACCEPT nutrition IGNORE food ACCEPT anti-microbials chemicals if linked to mp 3 or qualified as being poisonous DO NOT ACCEPT sebum ACCEPT stimulate the immune system (toxins) kill pathogens	(2)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	keratin / antimicrobial secretions (by the skin) / oils / sebum	ACCEPT barrier IGNORE sweat explanations and other methods	(1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A description that includes the following points: - majority of bacteria are three types - males of all ages have the same (predominant) groups of bacteria present on their skin (1) - Proteobacteria are the highest group of bacteria in all age groups (1) - middle aged men and elderly men have similar proportions of the three types of bacteria present on their skin (1)	IGNORE reference to 'others' throughout Do not piece together ACCEPT Firmicutes is lowest named bacteria in all groups ACCEPT <u>teenagers</u> have { <u>more</u> Firmicutes and Actinobacteria / <u>less</u> Proteobacteria} (than middle aged and elderly men)	(3)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A description that includes two of the following points: • collect from same part of {body / skin} (1) • men should use same washing regime (1) • men from same environment (1) • men should not be taking antibiotics (1)	ACCEPT description e.g number of showers, same soaps ACCEPT same parts of the world, named environmental conditions e.g. UV light, humidity, temperature ACCEPT should not use antibacterial {soaps / creams} no skin conditions	(2)

Question number	Answer	Additional guidance	Mark
3(c)(i)	A description that includes the following points: - macrophages {engulf / phagocytose / ingest} the bacteria (1) - and destroy the bacteria with enzymes (1)	ACCEPT pathogens / microorganisms DO NOT ACCEPT viruses DO NOT ACCEPT in context of opsonisation ACCEPT digestive enzymes / named digestive enzyme / lysozyme	(2)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	A description that includes two of the following points: - macrophages present antigen (on surface / on MHC) (1) (macrophages present antigen) to {T helper / CD4} cells (1) - macrophage engulf opsonised bacteria (1)	DO NOT ACCEPT to T killer cells / B cells / other cells ACCEPT description of opsonisation	(2)